

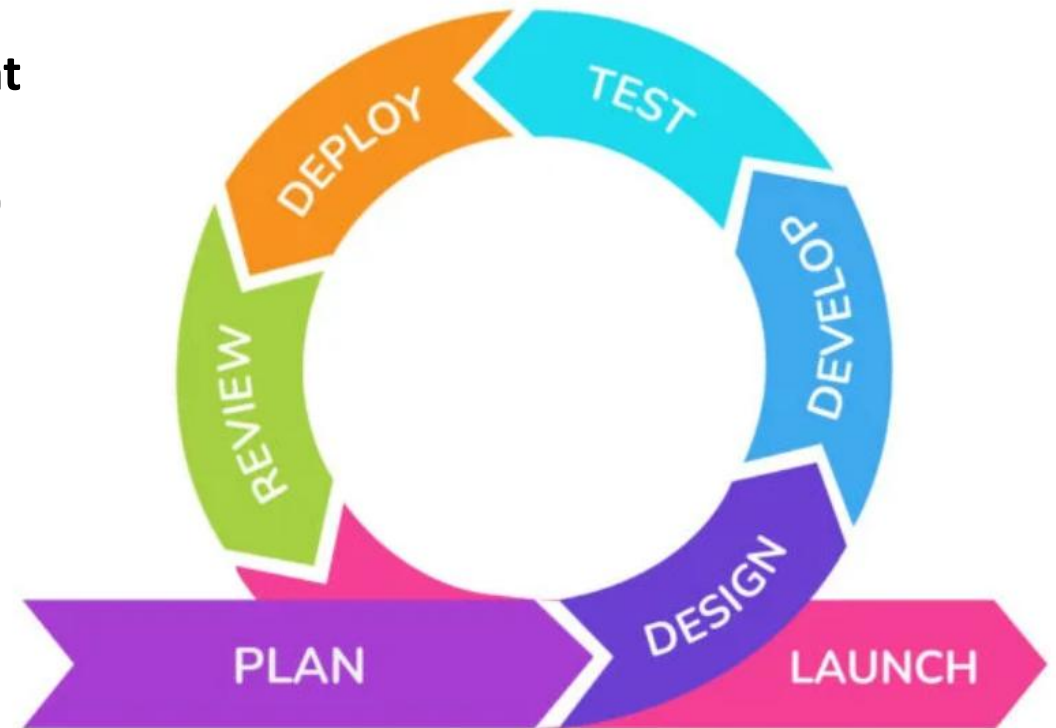
Microsoft Azure CI/CD Concepts

Introduction to CI/CD

AGILE

Agile is a **project management and software development methodology** focused on:

- Delivering work in **small, iterative cycles** (called sprints)
- Responding quickly to change
- Frequent feedback from customers
- Continuous improvement

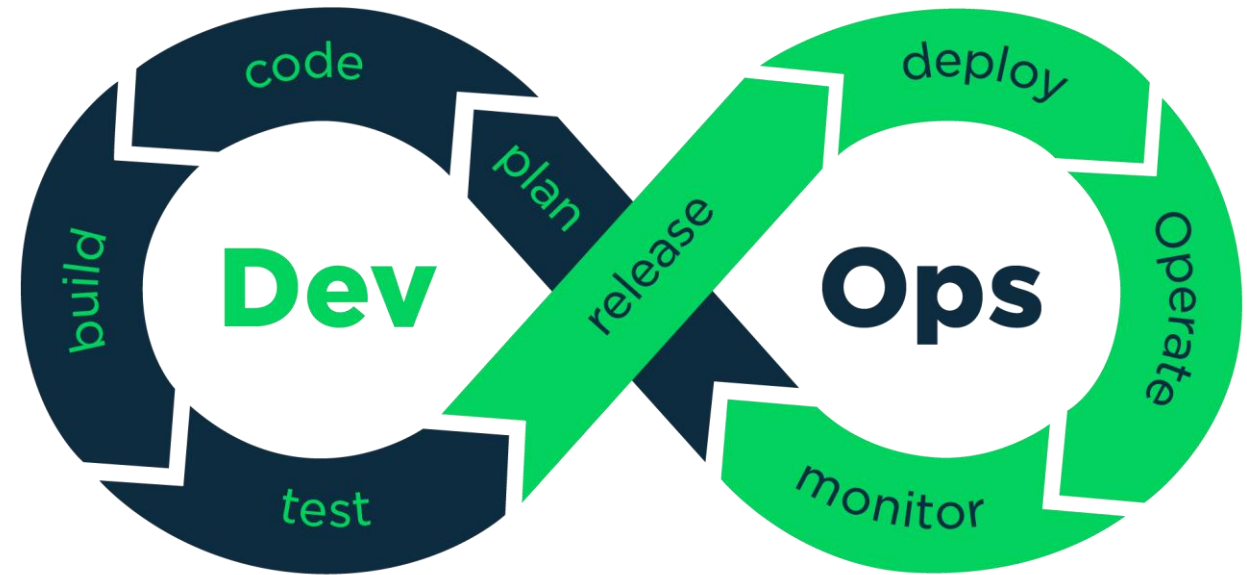


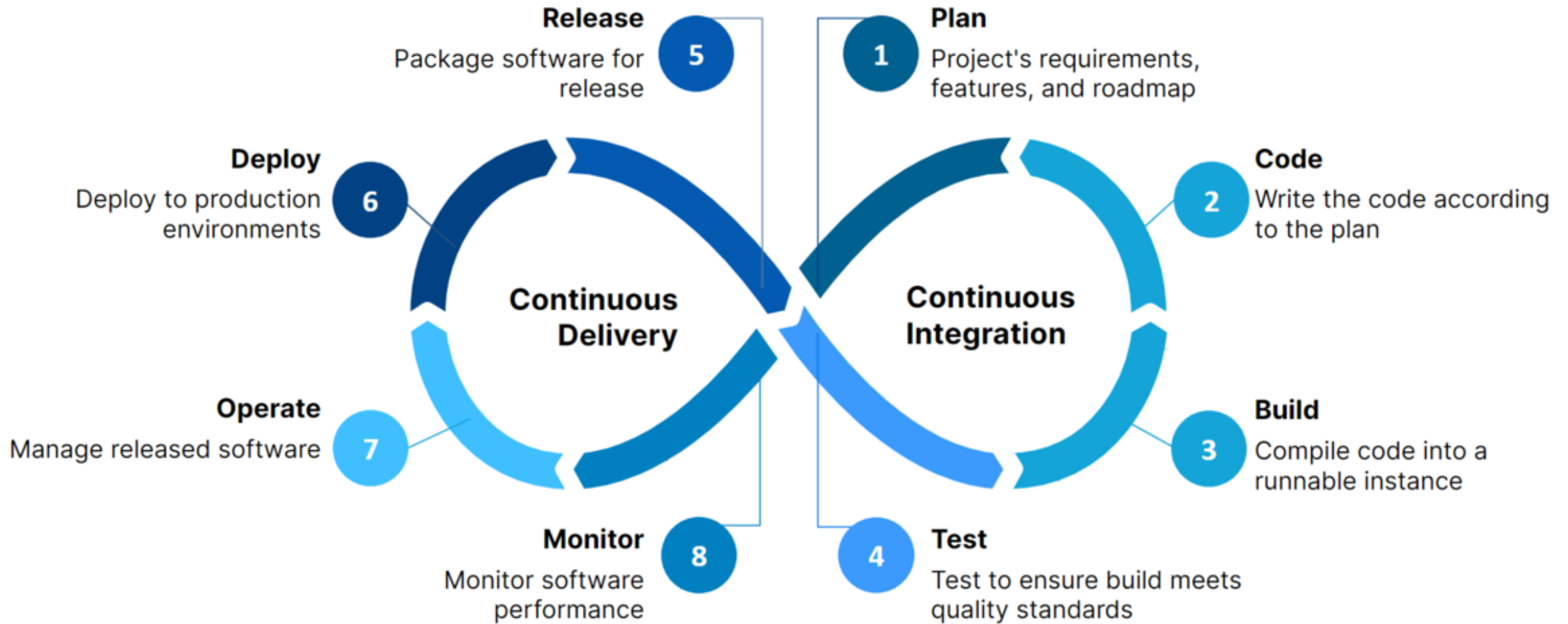
DevOps

DevOps is a **culture and set of practices** that bring **software developers (Dev)** and **operations teams (Ops)** together.

DevOps focuses on:

- Collaboration between teams
- Automation of processes
- Faster and more reliable delivery of software
- Monitoring and improving systems in production
- DevOps fills the gap between writing code and running that code smoothly in the real world.





1. Introduction to CI/CD Concepts

What is CI/CD?

- **CI (Continuous Integration)**

A development practice where developers frequently merge (“integrate”) their code into a shared repository. Each merge triggers automated builds and tests to catch issues early.

- **CD (Continuous Delivery / Continuous Deployment)**

Automates the release of new versions of the app.

- **Continuous Delivery:** Code is automatically tested and prepared for deployment, but deployment requires manual approval.
- **Continuous Deployment:** Code is automatically deployed to production with no manual step.

Benefits in Real-World Development

- Faster development cycles
- Fewer bugs and integration issues
- Higher code quality through automated testing
- Easier rollback and improved reliability
- Predictable deployments (less “deployment day” stress)

CI vs CD vs DevOps

Term	Simple Meaning
CI	Automatically build & test code whenever developers push changes
CD	Automatically release or deploy build artifacts to environments
DevOps	A culture + set of practices combining Dev + Ops to improve collaboration

2. Azure with GitHub

GitHub Webhooks

- A webhook is a mechanism that **notifies Azure** (or other systems) automatically when something happens in GitHub—like a commit, pull request, or new branch.
- It is used to **trigger CI/CD pipelines**.

Creating GitHub Jobs

- In GitHub Actions, a “job” is a group of steps that run on a virtual machine.
- Example jobs: build code, run tests, deploy to Azure.
- Jobs run in YAML files inside `.github/workflows` .

Pulling Code from GitHub

- CI/CD tools (like Azure DevOps Pipelines or GitHub Actions) fetch your source code using:
 - Git checkout actions
 - Built-in pipeline tasks
- This allows the pipeline to build, test, and deploy your code automatically.

3. CI for a .NET Core App

Build and Test .NET Core Backend

Common pipeline tasks include:

- **Restore dependencies:**

```
dotnet restore
```

- **Build project:**

```
dotnet build --configuration Release
```

- **Run tests:**

```
dotnet test
```

- **Publish artifacts:**

```
dotnet publish
```

Configuration for .NET Core CI

- Use a GitHub Actions YAML or Azure Pipeline YAML.
- Example configuration includes:
 - Select .NET SDK version
 - Restore packages
 - Build and test
 - Output artifacts for release
- CI ensures every change compiles and passes tests.

4. CI for a React App

Install Dependencies

Use Node.js and npm/yarn:

- `npm install`
- or
- `yarn install`

Run Tests

- `npm test -- --watch=false`
- Ensures UI logic works properly.

Build the App

- `npm run build`

Generates static files for deployment (HTML, CSS, JS).

Separate Pipeline or Stages

You can:

1. **Use separate pipelines** (one for .NET backend, one for React frontend)
 - Easier to manage
 - Independent deployments
2. **Use separate stages in the same pipeline**
 - Single YAML
 - Backend → Frontend → Deploy stages
 - Stages can depend on each other or run in parallel

✓ Summary Table

Concept	Simple Definition
CI	Automatic building/testing when code changes
CD	Automatic preparation and/or deployment to environments
Webhook	GitHub event → triggers Azure pipeline
GitHub Job	Scripted steps that run during CI/CD
.NET Core CI	Restore → Build → Test → Publish
React CI	Install → Test → Build
Separate Pipelines	Backend and frontend each have their own workflow
Stages	Multiple steps inside one pipeline

Q & A

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